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The perforator hypothesis: knots as small arteries held shut where they pierce the fascia

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Correspondence and objections: github.com/ejhong/knots/issues · PDF · the companion preprint

ABSTRACT

Anyone with a foam roller can find them: small, tender places in the soft tissue that persist for years yet can let go within seconds, sometimes with a tingle across a patch of skin. No mechanism for them is established, and no release has been recorded as it happens. We propose one. A knot is a cutaneous perforator, a small artery travelling with its veins and a nerve through an opening in the fascia, held shut by the smooth muscle of its own wall. Each part rests on measured physiology. A toned small artery can snap shut and stay shut until its tone falls well below the level that shut it; held shut for hours, it relaxes only slowly and incompletely; the breath moves the skin's sympathetic drive within seconds; a squeeze and release widen a small artery; a nerve deprived of blood fires when blood returns; and a vessel held shut holds shut the branches it feeds. A knot, then, is a memory with an anatomy: the held vascular contraction of Johnson's vasocomputation, placed in one class of vessel, written by the sympathetic drive, kept by the wall rather than by the latch of his account, and read by the perforator's own nerve. We argue that it is now the strongest account in a specific sense: it explains more of what is reported, with less that is new, than six alternatives in their strongest forms. We state what it does not explain, weigh the claim that acupuncture points sit on perforators, describe runaway release, a rare and dangerous process reported after forceful efforts to clear knots, with the slower changes in the fascia that follow it, and give the measurements that would refute it.

A caution. One process discussed here, runaway release (Section 5), can be prolonged and dangerous, and we think it can bring a person to the point of suicide (Sections 5 and 9). This paper describes; it does not prescribe. Nothing in it is a practice or a protocol, and no one should try to bring these experiences on. Anyone in difficulty can find support from Cheetah House; anyone thinking about suicide should reach out now: in the US, call or text 988; elsewhere, find a line at findahelpline.com.

1. Introduction

1.1 What is reported

Anyone with a foam roller can find them: small, discrete, tender places in the soft tissue, dozens of them, by many accounts densest around the hips and shoulders, and more with every decade. Massage therapists and physiotherapists call them knots or trigger points; contemplative traditions speak of knots in the body's channels. Pressed, a knot announces itself unmistakably. Stretched, it gives not the sharp pain of a muscle at the end of its range but a dull block, as though layers had stopped sliding past each other.

Those who work on knots, the author among them, describe a consistent set of features (Box 1). These are impressions gathered informally, not measurements. Some are echoed in systematic work: in a study of meditation-related challenges among Western Buddhist practitioners, pressure and tension that grew more acute or released in the course of practice were one principal set of the somatic changes reported [1]. We take Box 1 as what any account of knots must explain, and no more.

Box 1. What practitioners report

- They persist until released, for years if left alone, yet any of them can let go quickly. Some are more stubborn than others; freeing those takes more skill, above all with the breath, and the skill takes training. The toughest may take a skilled practitioner a whole session, never months.
- They let go within seconds, not days: under a hand, often as it lifts; under a roller; with the breath, by many accounts less with a single deep breath than with a subtle, complex pattern of breathing joined to fine attention to the body.
- Deeper ones let go to focused attention, sometimes without touch.
- Warmth helps: a hot shower softens many at once. Some find that a drink of water eases releases within about ten minutes, though on the account below it should not (Section 3.5).
- A release can come with a small pop, or a tingle across a patch of skin.
- They limit movement: a stretch meets a dull block at the knot, unlike the sharper pull of a tight muscle, and ordinary stretching does little to free it.
- They come in families. When one lets go, others nearby may go with it, or another may appear close by; a large one can free smaller ones around it, not the reverse.
- They gather where stress is held, and accumulate with age.

- Some come back to roughly the same place.

1.2 A state, not a structure

A knot that lets go within a breath cannot be a scar or an adhesion: what the body has built, it cannot unbuild in a breath. It must be a state the body is holding, as the first essay on these knots put it [2]: the tissue stays where it is, and the knot is a state written on it. The question is which state, held where, and by what. Several candidates have been proposed (Section 4). We argue for one, and claim for it something specific. At present it is the strongest account in three senses: it explains more of Box 1 than any alternative; it does so with physiology that is already measured, adding a new identification rather than a new mechanism; and it is the cheapest to refute.

1.3 Where the hypothesis came from

The hypothesis grew in public. The first essay [2] followed Johnson's vasocomputation [3], in which a held vascular contraction stores a state, and placed the store in a latch in the smooth muscle of small arteries inside muscle. The second [4] moved it to the perforators, the bundles that carry an artery, its veins and a nerve through the fascia to the skin, and supposed that a collar of gelled hyaluronan around each held it; a longer preprint of the same month set out that version at length [5]. Checked against physics, the collar failed. At its most viscous, hyaluronan around a vessel pushes back about 12 times more weakly than the blood pressure holding the vessel open [6, 7, 8], and on its own, in physiological conditions, it stays a liquid rather than gelling [9]. If anything holds a perforator shut, it is the wall. That is the form argued here; where the earlier versions differ, this one replaces them. It remains a case of vasocomputation's idea, not a rival to it. Johnson proposes that a held vascular contraction stores a state, that the state is a prediction, and that the contraction clamps the neural circuit around it. The hypothesis takes the first proposal and places it in one class of vessel; it needs neither of the others, and contradicts neither. Where it departs from Johnson is in what makes the state last: in his account the latch, holding a contraction cheaply for as long as its prediction stands; here the wall's own mechanics, which keep a vessel shut after the drive that shut it has eased and, once it has set, at rest (Sections 3.2, 3.3 and 4). A companion preprint runs seven candidate mechanisms evenly, as models [10]; this paper argues for one.

2. The hypothesis

> *A knot is a cutaneous perforator, a small artery travelling with its veins, a nerve and a lymphatic through an opening in a fascia, held shut by the smooth muscle of its own wall. It lets go when the wall's tone falls, or the wall is moved, enough for the vessel to reopen; blood returning to the patch the vessel feeds wakes its nerve.*

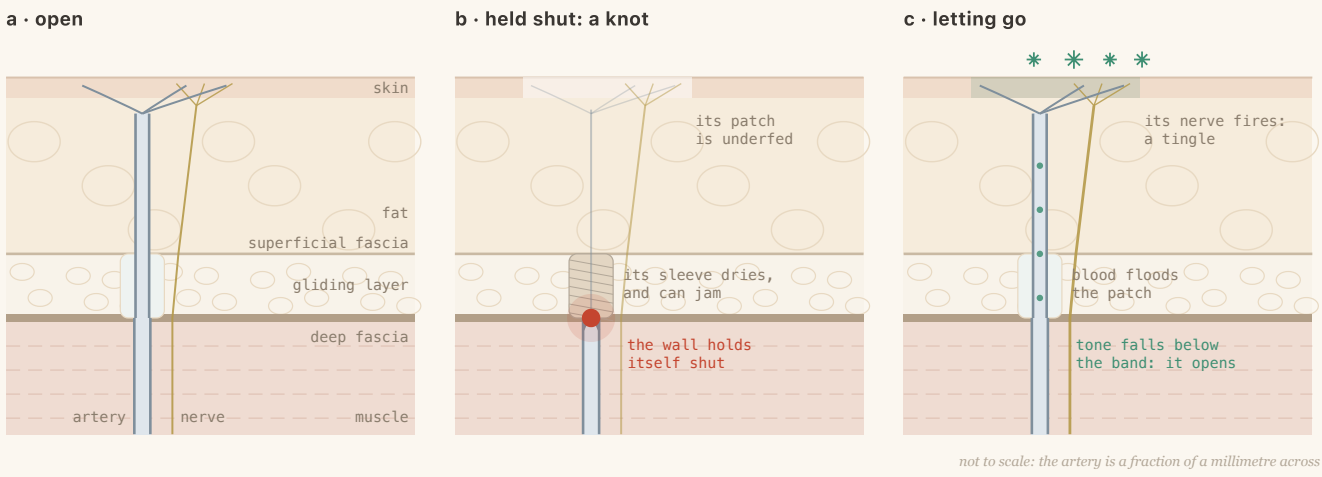


Figure 1. One perforator in section. (a) Open: a small artery and its nerve rise through an opening in the deep fascia, cross the gliding layer in a sleeve of sliding tissue, and branch in the skin. (b) Held shut by its own wall at the opening: a knot. Its sleeve dries and can jam; the patch it feeds is underfed. (c) Letting go: tone falls below the band that held it, blood floods the patch, and the nerve fires.

In parts, each of which can be checked on its own:

1. **Place.** Knots sit where perforators pass through the deep fascia (the larger ones) or the superficial fascia (the smaller ones), at fixed sites. · new
2. **Hold.** A small artery with tone in its wall can close and stay closed until its tone falls well below the level that closed it. · established
3. **Set.** Held shut for hours, the wall's muscle adapts to its shortened length and relaxes only slowly and incompletely; over days the artery remodels inward. The vessel could then stay shut nearer to resting tone. · established in small arteries; in knots, new
4. **Drive.** What carries a vessel shut is the sympathetic drive to the skin's vessels, raised by stress and moved within seconds by the breath. What lets it go is that drive falling, warmth, or the wall being moved: shaken, or squeezed and released. · established physiology; its role in knots, new
5. **Nerve.** While the vessel is shut, the patch it feeds is underfed and tender; when blood returns, the nerve that went without it fires: the tingle of a release. · the tingle established; the tenderness, new

6. **Sleeve.** Each vessel crosses the gliding layer between the fasciae in a sleeve of sliding tissue that its flow keeps wet; a vessel shut for long lets its sleeve stiffen, pinning the layers there: the dull block. · `hypothesis`
7. **Trees.** Perforators branch from parents. A shut parent holds its children shut, and when it opens they can open with it; a child that opens cannot open its parent. · `hydraulics; in knots, new`

A memory with an anatomy

Read as a whole, the hypothesis describes a memory with an anatomy. In Johnson's account a held vascular contraction acts as a medium-term memory [3], and the essay that brought the idea to the perforators asked what would write, store and read such a state [4]. Here the answer is concrete. The sympathetic drive writes: stress raises it, and a perforator it carries shut stays shut as the drive eases, so long as its tone does not fall below the band; only adaptation and remodelling let it hold at rest. The writing is local. Vessels differ in the thickness of their walls, the pressure that feeds them and the load at their openings, so a rising drive carries a few of them across the band at a time, not all at once. The wall keeps: its switch holds the state, and adaptation and remodelling make it durable. The perforator's own nerve reads: as tenderness while the vessel holds, and as a tingle when it lets go. It reads through the blood the vessel withholds and returns, one of the ways Johnson expects a contraction to act on the neurons nearby [3]. Each perforator is a site that can be set or cleared, and a body has hundreds of large ones and many more small ones [11]: a map of held states that a hand can find, the breath can reach and an instrument could read. In this sense the hypothesis gives one of vasocomputation's memories an address; whether what it stores is also a prediction, as Johnson proposes, it leaves open.

3. The case

No single observation below proves the hypothesis. The argument is that one anatomical object, behaving as measured physiology says it behaves, accounts for nearly every feature in Box 1, where each alternative accounts for some.

3.1 Where knots are

Perforators are discrete, numerous and fixed. Taylor and Palmer's whole-body injection studies plotted an average of 374 major perforators in each body, the arteries closely following the connective-tissue framework [11]. They pierce the deep fascia at sites that recur from person to person: the clusters are shared, but the individual vessels scatter around their cluster's centre, in the leg with standard deviations of 6–11% of the tibia's length, roughly 2–4 cm [12], and differ between the two sides of one body [13]. That is how knots are described: in the same regions in everyone, never at quite the same point.

They are also where the layers are held. The superficial fascia is tied to the deep fascia by fibrous bands, the retinacula cutis, which work as tethers and anchors [14], and perforators cross the same gap, large and sparse where the skin is mobile, smaller and denser where it is tethered [15]. Pressure from above presses each bundle against the firm deep fascia beneath it: a hand or a roller meets a perforator where it can squeeze it.

And perforator exits are already used to find tender points. In entrapment of the cutaneous nerves of the abdominal wall, the perforating vessels' exit through the rectus fascia beneath the point of maximal pain is found by Doppler and used as the landmark for injection [16]; lidocaine injected at the point of maximal pain relieved it in 13 of 24 patients, saline in 4 of 24 [17]; and a case report describes a tender point with a pulsating perforating artery directly beneath it on Doppler, after whose release with fluid the sign placing the pain in the abdominal wall disappeared and pain and movement improved [18]. These implicate the nerve as much as the vessel. They show that the address is right.

3.2 What holds them

A small artery is a balance between the pressure inside it, pushing it open, and the tension of its muscle, pulling it shut. Burton showed that with active tension in the wall the balance has no stable open state below a critical pressure: the vessel snaps shut [19]. Whole vascular beds stop flowing at such a critical closing pressure [20], and failing skin flaps show it [21]. Because a muscle is weak when it is very short [22], a vessel that has closed needs far less tone to stay closed than it took to close it. Over a band of tone, open and shut are both stable, and a shut vessel reopens only when its tone falls below the band, when outside pressure is relieved, or when the pressure feeding it rises [23]. In living tissue, nerves hold some small arterioles fully closed at rest: in the muscle of anaesthetised rats, cutting them opened 22% more [24].

With the measured strength and wall thickness of human small arteries [22, 25], the project's model of the wall finds that band between 0.62 and 0.93 of maximal tone, with rest at about 0.19 (Figure 2). Across 49,152 plausible parameter sets, about half (52%) give the switch, and only 6% hold shut at resting tone [8]. The hold is possible but demanding: a healthy vessel holds itself shut only while its tone stays well above rest. Thicker walls, as in hypertension [25], make it easier.

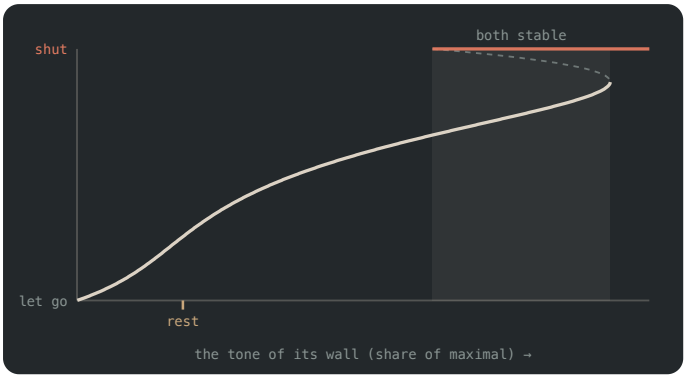


Figure 2. The switch in a small artery's wall, from the project's model with measured parameters: how far shut the vessel is (up, from relaxed to shut) against the tone of its muscle (across). Over the shaded band, open and shut are both stable. Tone rising past the band snaps the vessel shut; once shut, it reopens only when tone falls below the band.

3.3 How knots form, and why old ones persist

How a knot forms follows from the band. Shutting a vessel takes a surge: in the project's model, with measured parameters, the tone that snaps it shut is about 5 times its resting tone, and a deep breath at rest, sized to the fall in fingertip flow it causes (Section 3.4), shuts no vessel in any of the 49,152 plausible parameter sets [8]. A fright, pain, cold or a squeeze can: local cooling all but stops the skin's flow [26], and pressure from outside lowers the tone it takes [23]. Staying shut takes less, but the stress must last: a constriction of five minutes leaves no trace [27, 28], while one of hours begins to set. A knot, on this account, is the trace of a coincidence, a surge and a stress that outlasted it; a brief stress leaves none. That may be why knots gather where stress is held for long, and why children have few.

Smooth muscle adapts to the length at which it is held. Arterioles constricted for five minutes relax fully when the drive is removed; after four hours they relax slowly [27] and incompletely, their cells repositioned to hold the smaller diameter [28], though the hold still needs calcium and goes at once without it [27]. Smooth muscle held short regains much of the force it lost there [29, 30], and resistance arteries kept constricted for three days remodel inward, their relaxed lumen 10–16% smaller [31]. A vessel held shut long enough might then hold nearer to rest: the knot would set, and the setting would deepen with time, while still answering to whatever lowers the wall's calcium. A set knot would be more stubborn: holding at rest, it lets go only when the drive falls well below rest, in the model when the wall's tone falls to between 33% and 51% of its resting level [8], and the slower relaxation of arterioles held for hours suggests it may also take longer to go. The latch, by which smooth muscle holds force at less cost [32], may be at work in the wall as well, but it would make the hold cheaper, not longer: relaxation is paced by the fall of calcium, not by the latch [33]. Practitioners describe knots that differ in how stubborn they are, any of which a trained breath can free quickly; on this account, that asks the breath to take the drive, or the wall's calcium, well below rest (Section 8). The setting also predicts something an isolated vessel can show (Section 7).

3.4 What the breath can reach, and how fast

The skin's small arteries are governed by sympathetic nerves: skin blood flow runs from almost nothing under local cooling to several litres a minute in heat [26], and about a third of the superficial fascia's nerve fibres are autonomic, gathered around its vessels [34]. The breath reaches this drive within seconds. A deep inspiration constricts the skin's vessels [35], cutting fingertip flow by 59–71% [36], beginning 3.6–3.8 s after the breath, deepest at about 5 s, and recovering over the following ten to thirty seconds [37]. Slow breathing works the other way: fifteen minutes of it lowered sympathetic activity to muscle by five bursts in every hundred heartbeats [38]. A single deep breath, then, briefly raises the drive to the skin's vessels, which may be why releases are reported less with one deep breath than with a subtler pattern (Box 1). These measurements come from the skin of the fingers and toes, unusually rich in sympathetic control, and from the nerves to muscle; whether the breath moves the drive to the perforators of the trunk as fast, or as far, is not known. Movement is a possible second route: vibration cuts the active force of vascular smooth muscle at once [39], but at 1 to 400 cycles a second, far faster than a breath, and whether the slow stretch of breathing does the same is untested. Which route matters, and which pattern of breathing serves it, is open; the hypothesis requires only that a releasing breath lower the drive more than it raises it, or move the wall.

3.5 Why hands, rollers and warmth

A vessel pressed shut from outside cannot reopen while the pressure lasts; it can reopen as the pressure lifts [23]. A knot under a hand should therefore let go as the hand lifts or eases, not while it presses. Squeezed and released, the feed arteries of rat muscle widen within seconds, and several brief squeezes widen them more than one long one [40]: a roller's passes are a series of brief squeezes, and it shears the tissue around the vessel as well. Heat acts more slowly and more widely. Local warming raises the skin's flow most of the way to its maximum [41], but a patch of skin warms with a time constant of about 44 s even at maximal flow [8]: too slow to free a knot within a breath, fast enough over the minutes of a hot shower, and everywhere the water touches.

Some also find that a drink of water eases releases within about ten minutes (Box 1). On this account it should not. Drinking water raises the sympathetic drive: in healthy people the nerves to muscle fire more, most at half an hour, and the calf's vessels narrow [42], and in young men half a litre lowers the skin's flow in the palm at twenty-five minutes [43]. Systolic pressure, which would help a shut vessel open (Section 3.2), rises by about 11 mmHg in older people and not in the young [44]. A narrower vessel in the skin is a harder knot to free, so in the young, whose pressure does not rise, the report counts against the hypothesis, unless the drinker was short of water: after exercise that dehydrates, resting blood pressure

is lower, and fluid given by vein lowers the drive to muscle [45]; whether a drink does the same is not measured. A recording would tell (Section 7). Nor is water harmless in quantity: more than five litres drunk within a few hours has diluted the blood's sodium enough to swell the brain, and has killed [46].

3.6 Why a knot is tender, and why a release tingles

While a perforator is shut, the patch it feeds is underfed, and so is the nerve that travels with it. Underfed, not starved: a perforator's territory is linked to its neighbours' by anastomoses [47], through which some blood can still come. At active trigger points the tissue is acidic and rich in the substances that sensitise pain nerves [48]; an underfed patch of skin and fascia could build the same chemistry, and its nerve would be tender to pressure. On this account a knot is tender because it is shut. When flow returns after a five-minute arterial occlusion of the forearm, the skin's flow overshoots, rising by about 187% and peaking about 11 s after the release [49], a response in which sensory nerves and BKCa channels take part [50]. A nerve that has gone without blood also fires on its own when the blood returns: recorded in human sensory nerves, this firing begins about half a minute after circulation is restored and peaks by the second minute, as the felt tingling does [51]. When a shut perforator opens, blood floods its patch and the nerve that went without it fires: on a small scale, the prickle of a limb waking after pressure. The tingle should follow the opening within about a minute and cover the vessel's territory, a patch of skin rather than a line [47]. Two cautions. Overbreathing increases such firing [51], so a tingle during forceful breathing is weak evidence by itself; and a single perforator's flood is far smaller than a cuff's, which is why a recording needs a pressed sham beside it (Section 7).

3.7 Why a dull block

Each perforator crosses the gliding layer between the superficial and deep fasciae, a web of small fluid-filled cells that lets skin, fat, fascia and vessels slide past each other [52]. The vessel's flow keeps its stretch of this web wet. Shut for long, the sleeve would dry and stiffen. This is not the collar of Section 1.3, which had to hold a vessel shut against the blood pressure and could not. The sleeve need only resist the small shear of layers sliding past each other, and for that it needs a material that holds until pushed past a threshold. The gliding layer offers candidates: crowding and immobility raise hyaluronan's concentration and viscosity and reduce the gliding of the layers, and in inflammation modified hyaluronan can form aggregates or gel-like structures [53]; materials of this kind can jam below a critical stress and give way abruptly above it [54]. A jammed sleeve pins the layers at that point, and a stretch meets a block there. Fascial layers glide less in chronic low back pain [55], and the loose sublayers of the neck's fascia are thicker in chronic neck pain [56]. The sleeve is the least established part of the hypothesis, and it predicts an order a recording could check: the flood and the tingle as the vessel opens, the sliding only once the layer is moved.

3.8 Why releases come in families

Perforators form trees: a parent vessel feeds children around it. A child cannot flow while its parent is shut, so a shut parent holds its children shut, and when the parent opens, those not held by their own walls open with it. A child that opens does nothing for a parent that stays shut. The asymmetry is hydraulic: pressure comes from the source. Signals along the wall would not supply it, for a dilation started on an arteriole spreads both ways [57], fading with a length constant of about two millimetres, and those started on daughter vessels sum in the parent [58]; over the centimetres between one perforator and the next, they would have faded. Release should then run in families: freeing a large knot frees small ones around it, and freeing a small one does not free the large one. Trigger-point practice describes a similar order: dry needling a key trigger point raised the pressure-pain threshold of satellite points in its zone of referred pain [59], though a spinal mechanism has been proposed for it.

3.9 Why more with age

Small mesenteric arteries from organ donors over sixty-five contract more strongly to adrenergic stimulation [60], and resting sympathetic activity to muscle rises with age [61]; older skin reaches a lower maximal flow when heated [62]. All three would make vessels likelier to hold and harder to flood open. But the skin's reflex vasoconstriction is blunted with age [63], so whether the drive to the skin's own vessels rises is not settled. The hypothesis predicts that the number of knots tracks the loss of the skin's microvascular reactivity, place by place, and that children have few.

Table 1. What the hypothesis accounts for, what it rests on, and how firmly.

Reported	The perforator account	Rests on	Status
Discrete, tender places in the same regions in everyone	Perforator exits: fixed sites, shared clusters; tender because shut	[11, 12, 16, 48]	Identity untested
They persist until released; some are stubborn	A wall that holds itself shut; muscle that adapts; an artery that remodels	[19, 28, 31]	Each measured; together untested
They let go within seconds	A shut vessel reopens as soon as its tone falls below the band	[19, 20, 23]	Established mechanism
With the breath	A deep breath raises the drive for seconds; slow breathing lowers it	[37, 38]	Measured in fingers and in nerves to muscle; in knots, untested
As a hand lifts; under a roller	A pressed vessel reopens only on release; brief squeezes widen it	[40]	Measured in feed arteries of rat muscle
A hot shower frees many	Local warming lowers the tone of the skin's vessels	[41]	Established

Reported	The perforator account	Rests on	Status
Water eases releases (some)	Drinking narrows the skin's vessels; it could help only someone short of water, or whose pressure it raises	[42, 43, 45]	Against, unless short of water
A tingle over a patch	A nerve that went without blood fires when it returns; reactive hyperaemia	[50, 51]	Established
A dull block to stretch	A dry sleeve jams in the gliding layer	[52, 53, 54]	Hypothesis
Families; a large one frees small ones	Trees: a shut parent holds its children shut	[11]	Hydraulics; in knots, untested
More with age	Stronger adrenergic contraction; less capacity to dilate	[60, 63]	Mixed
Deeper ones go to attention alone	No direct route	—	Not explained

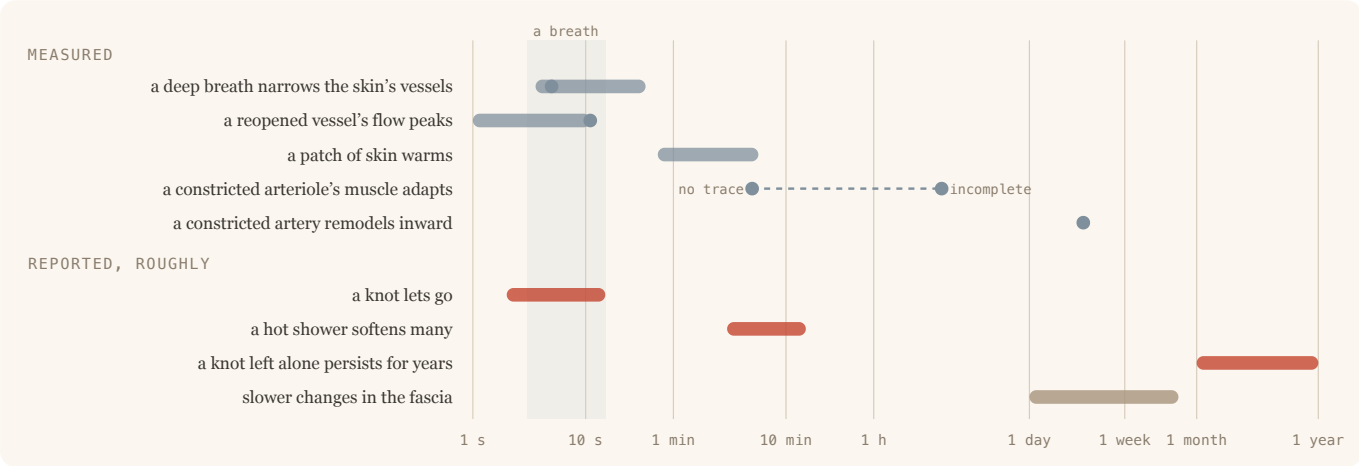


Figure 3. The clocks. How fast knots are reported to change (terracotta; rough, from Box 1), beside how fast the physiology the hypothesis names is measured to change (slate). A deep breath narrows the skin's vessels from 3.6 s, most at about 5 s, recovering over the following ten to thirty seconds [37]; a reopened vessel's flow peaks about 11 s after release [49]; a patch of skin warms with a time constant of 44 s at maximal flow and 341 s at a tenth of it [8]; arterioles constricted for five minutes keep no trace of it, and after four hours relax only slowly and incompletely [27, 28]; arteries constricted for three days remodel inward [31]. The slower changes in the fascia (Section 5) are reported over days to weeks. The shaded band is a breath.

4. The alternatives

Six other mechanisms have been proposed or can be assembled from established physiology (Table 2). Each has a strongest form, and most have evidence the perforator hypothesis lacks. The closest is the vascular latch: Johnson's account of how a held state lasts, and the form from which this hypothesis grew. It is the same muscle, held by a prediction and its latch rather than by the wall's mechanics; the two part in what lets a knot go and in what its release would show.

Table 2. Six alternatives in their strongest forms, and the trace each would leave at the moment of release.

Mechanism	What holds a knot	Best evidence	Explains, and struggles with	At release
Trigger points	An energy crisis: sustained contraction squeezes its own capillaries, and the ischaemia withholds the energy needed to relax [64]	Stiff nodules on ultrasound [65]; an acidic milieu [48]; spontaneous electrical activity [66]; lactate rises after compression [67]	A palpable nodule, a twitch, referred pain. It struggles with release in seconds to the breath and with a tingle over the skin; finding points by touch is only moderately reliable [68], and the construct is contested [69]	Stiffness falls in the muscle

Mechanism	What holds a knot	Best evidence	Explains, and struggles with	At release
Densification	Hyaluronan between fascial layers grows viscous, and in inflammation can form gel-like aggregates; the layers stop gliding [53, 70]	Thicker loose sublayers in neck pain [56]; less glide in back pain [55]	The block and the slide, and perhaps the slower changes in the fascia (Section 5). It struggles with seconds: hyaluronan on its own stays liquid [9], and warmth thins it only two- to threefold over forty degrees [53]	The layers slide again
Nerves	A cutaneous nerve sensitised where it pierces the fascia [71]	Established entrapments [17]; fibres that fire when pressed after a neuritis [72]	Tenderness, and a local anaesthetic test. It has no hold of its own, and in one preparation, in nerve-injured rats, sympathetic stimulation changed sensory firing only after about fourteen seconds [73]	The nerve's firing falls
Vascular latch	A contraction held long enough engages smooth muscle's latch and freezes a prediction in place, cut off from awareness and from updating [3]	An integrative account; untested	Release to attention alone. The latch saves energy but does not by itself remember [33]	Flow rises slowly
Motor switch	Motor units, once recruited, keep firing on persistent inward currents [74, 75]	Self-sustained firing measured in people	Release to attention and inhibition, at once. It struggles with a tingle over a patch of skin, and with a release at a site with no muscle beneath it	A motor unit falls silent
Perception	Central gain, kept up by threat, attention and poor sleep, makes ordinary input felt as a knot [76]	Central sensitisation is established in chronic pain	Variability and global effects. It struggles with discrete places that return to the same spot	Nothing changes in the tissue
Perforators	A small artery's wall holds itself shut	Its parts are measured; the identity is not	Most of Box 1; not release to attention alone	Skin flow floods in

Three points about the comparison. First, the perforator and the nerve share an address. A cutaneous nerve passes through the same opening, so finding tender points on perforators would not separate them; low flow before a release, and a flood at it, would.

Second, the perforator hypothesis has no direct route for knots that let go to attention alone, without touch. The latch or the motor switch would; so would a perforator whose tone is held in its band by a held prediction, the latch's command acting on the perforator's wall. That would join the two accounts: a held prediction keeps up the drive, the wall keeps the vessel shut, and the knot lets go when attention lets the prediction update.

Third, the alternatives need not all be rivals. Knots may be more than one thing, and densification, which struggles with release in seconds, may describe the slower changes of Section 5. A recording should classify each release rather than crown one theory.

The companion study writes all seven mechanisms as models from their own physiology and runs them through the same scenes [10]. We give its comparisons little weight here: most numbers specific to knots are guesses there, the formalisations of three of the seven are the project's own and have not been checked by those who hold them, and a shared calibration gives every model knots by construction. What the case rests on is the physiology above.

5. Beyond the knots: runaway release and the fascia

The knots are not the end of what is reported. Some who work forcefully at clearing them, in long and intensive contemplative practice or in somatic work, describe a rarer process, which we call runaway release. Past some point, releases come in chains, at times faster than they can be followed, and the process can no longer be set down: it feels unfinished, and stopping is harder to bear than going on. Each stage has to be found a way through, with a skill learned along the way. In the fullest accounts we know, knots go in bursts that spread over a region, or seem to travel before they let go. The layers of fascia then peel apart along winding lines, often with the body turning and twisting; places are left hollow, the hardest part, at times with nausea; and over days to weeks they fill in again, felt as re-stitching, which brings back a sense of wholeness. In any one region the knots go first and the layers after, while across the body the stages interleave. The whole course can run for months or years, and its last filling-in can be slow, uncomfortable rather than painful. People do come through it; those who have say it is lived through rather than stopped. The reports are many but informal; the process is little known, and we know of no study of it as such. It belongs to a wider record of bodily experience in intensive practice, which includes pressure and tension that build and release, currents of energy and involuntary movement [1], and energy-like somatic experiences whose course, effects and interpretation have been described in their own right [77]; we think some of what that record describes is this process.

Runaway release is arduous, and it is dangerous. Difficulties of this wider kind have lasted years and have included suicidality and stays in hospital [1] (Section 9). We think runaway release can bring a person to the point of suicide, above all someone facing it alone, exhausted, or without the experience, guidance and support to get through it. We describe it because people are beginning to work forcefully at clearing knots, and a risk that is known can be avoided, recognised and met with help.

Why releases would come in chains is clearest in a network. Each knot is a switch that holds until its drive falls below a band, and knots hold one another up: a shut parent holds its branches shut (Section 3.8), and an underfed, tender patch may add to the drive that keeps others shut. Clearing enough of them can lower the drive that holds the rest, so that each release makes the next easier, and past a threshold the network tips: releases run on until few are left. A postural or nervous-system set held up by its own inputs would tip in the same way (Section 4), so the tipping does not decide what a knot is.

The perforator hypothesis bears on the slower phase because perforators are among the things that hold the fascia down. The superficial fascia runs unbroken over the body and glides over the deep fascia on a plane of loose tissue [52, 78]; beneath a firm, protective superficial layer lies a mobile one, made for gliding [79]. It is held in place by fibrous tethers [14] and by the perforators that pass through it, each in its sleeve. If a shut perforator's sleeve jams (Section 3.7), each knot is also a point where the layers are pinned. Were many to let go across a region, and the returning flood to wet the gliding plane, the layers could part along the lines between the remaining tethers, and close again as the fluid is taken up and the sleeves hold once more. This is a hypothesis built on a hypothesis, and it is not yet modelled.

The hypothesis offers a second reading of the slower phase, in the vessels themselves. When the flow through small arteries rises and stays up, they remodel outward: in rats, an enzyme that breaks down the wall's matrix is more active by the fourth day, and by the fourteenth the arteries are wider and their walls have grown [80]; in mice, the remodelling needs macrophages, and the vessels' tone sets its direction [81]. Perforators that had long been shut, opened together, would carry such flow, and a clearing-out and rebuilding of their walls would run on about the clock the reports give for hollowing and re-stitching. This, too, is untested.

There is a strong alternative. Intensive practice brings long hours of attention to the body, and often overbreathing, strain and short sleep, each of which can produce vivid bodily sensation with no change in the tissue; attention to the body is not always benign, and past some point its effects can turn negative [82]. A reading in which these experiences are changes in how the body is perceived, the perception account of Section 4, is the natural first reading for a clinician. Practitioners in difficulty found grounding, calming and embodying activities, sleep and exercise helpful; body-based treatments, massage and acupuncture among them, were reported as helpful by some and not by others [1].

The two readings predict different things, and ultrasound can tell them apart. The glide between fascial layers can be measured [55]. If the layers are changing, glide should differ at the sites where the changes are reported, follow the knots rather than precede them, and settle over the days the reports give; if only perception is changing, the tissue should look as it did. On the first reading, densification, which cannot explain release within a breath, may be right about this slower family: hyaluronan in the gliding layer is where slow change in the fascia would live [53, 70].

6. Acupuncture points

Chinese medicine placed points along channels, and also took the tender spot itself as a point, the *ashi* point. If knots are perforators, it is natural to ask whether the classical points are too. Heine reported that about 80% of acupoints lie where a vessel–nerve bundle perforates the fascia [83]; the figure is widely repeated, but its method cannot be checked from the record. Dissection beneath 201 points found a vessel, mostly a vein, at about seven in ten, but no one structure common to all [84], and a multicentre dissection of four bodies and two lower legs found vessel–nerve bundles at only a few points [85]. Most points lie on the connective-tissue planes between muscles [86], which is where many perforators run. The one study we found with control points detected perforators by thermography at 87.5% of 144 lower-limb points and at 18.1% of matched control points, in six people, with no blinding described [87]; a dissection of two arms found a perforating vessel at all 30 points studied [88]. Control points a few centimetres away are not a measure of chance where perforators cluster, and none of these studies asked how often a point placed at random would land as near a vessel. Both maps are also fuzzy: 71 doctors' marks for one point spread over as much as 41 cm² [89], and perforator clusters spread by centimetres between people [12].

A first look against chance used five limb lines where a measured perforator cluster and a run of WHO points share an intermuscular septum [90]. The perforator density at the 21 points was about 1.23 times that at random points on the same lines, not reliably above chance (p 0.03–0.40 across 144 variants of the unstated details), where a map placed on the clusters would score 1.8–3.0 times. Some points sit on clusters, SP6 near the lowest posterior tibial perforator [91] and LU6 and LU7 near the radial artery's two clusters [92]; others do not. The correspondence is therefore not shown, and not excluded, and the hypothesis does not depend on it. Its nearest kin in the traditions is the *ashi* point, defined by tenderness alone.

7. Predictions, and what would refute them

Table 3. What the hypothesis predicts, how to look, and the result that would refute it.

Prediction	How to look	Refuted if
1. Tender points sit on perforators	Blinded palpation against a blinded Doppler or thermal map; distances against chance at several radii, with random and shifted maps	No excess over chance
2. A person has fixed stations	One person's knots marked over months against their own perforator map	New knots at arbitrary places, or continuous drift
3. Low flow before a release	Laser speckle over a knot's patch against its neighbours	Flow no lower than nearby
4. A flood at release	Laser speckle or a thermal camera at each felt release, with a pressed sham beside it	Release with no local rise, or a rise no larger than at a pressed sham
5. Release as pressure lifts	The same, with calibrated pressure	Releases under steady pressure, with flow rising while compressed
6. An order: flood, tingle, then sliding	Laser speckle, a button at the felt release, ultrasound of the layers sliding	Sliding returns before the flood
7. Families	Freeing an upstream site against a downstream one	Small frees large as often as large frees small
8. Setting	Isolated small arteries carried shut for four hours against five minutes, then returned to resting tone	Those held for hours reopen at rest as fully as those held for minutes
9. Age	A census of knots, with the skin's microvascular reactivity at the same sites, across ages	No relation; children as knotted as adults
10. The fascia follows the knots	Ultrasound of the glide between fascial layers in people who report the slower changes, at the reported sites and matched sites, over days	No difference from matched sites
11. Water does not help	Felt releases, laser speckle over knots and blood pressure for half an hour after half a litre of water against a sip, in people not short of water	Releases come more easily after the larger drink while the skin's flow falls and the pressure holds

The first is the cheapest and decides the most: an afternoon with a pencil Doppler or a thermal camera, two blinded examiners and a statistician. The fourth is the most direct: a flood of skin flow, time-locked to a felt release and absent at a pressed sham. Thermal imaging after a mild cold challenge already finds perforators, its first hot spots lying over vessels that Doppler hears and CT angiography shows [93], and laser speckle imaging of skin flow is reproducible from week to week [94]. Because each alternative predicts a different trace at the moment of release (Table 2), one careful recording would classify releases rather than merely confirm one view. None of these studies asks anyone to seek the experiences of Section 5: each observes what already happens, at the usual pace. A one-page pilot is set out separately [95].

8. Limits

- The identification is untested. No one has yet mapped tender points against perforators blind, or recorded a release.
- The reports of knots are informal, and partly the author's own.
- Finding tender points by touch is only moderately reliable from one examiner to another [68], so any map of knots must be made blind to the map of vessels (Prediction 1).
- A finger cannot feel a small artery shut beneath the fat. The hypothesis accounts for tenderness and the block, not for a firm nodule; where a nodule is found, it belongs to muscle, and that knot is not (only) a perforator.
- The hold is demanding in healthy vessels. That old knots hold at rest leans on adaptation and remodelling measured over hours and days in isolated small arteries, which relax incompletely rather than not at all. That a trained breath frees even stubborn knots quickly asks it to take the drive well below rest, or to move the wall, and no one has measured either.
- Water, which some find eases releases, narrows the skin's vessels in healthy people; on this account it should not help, except someone short of water or whose blood pressure it raises (Section 3.5).
- The evidence on age points both ways. The breath's effects are measured in the skin of the fingers and toes and in the nerves to muscle, not at the trunk's perforators, and which of its routes matters is unknown.
- Releases to attention alone are not explained.

- Runaway release and the slower changes in the fascia (Section 5) rest on informal reports, the hardest to check, and are the most speculative part of the paper.
- The text was written by AI (below). Every reference was taken from its own record, and the numbers from the cited sources and the project's data, but errors of emphasis or reading are possible, and corrections are welcome.

9. Safety

Meditation and related practices can harm. Across 83 studies, adverse events were found in about 8% of those who meditated, anxiety and depression most often, sometimes in people with no previous mental-health problems [96]. In the Varieties of Contemplative Experience study, which recruited practitioners who could describe challenging or difficult experiences, symptoms lasted a median of one to three years, from a few days to more than ten; nearly three-quarters reported moderate to severe impairment in some area of life, about one in six suicidality, and as many a stay in hospital [1]. From the same kind of sample, meditators in the West reporting challenges, the energy-like somatic experiences to which the reports of Section 5 belong have been described with their trajectories and their effects on people's lives, and with the many ways practitioners, teachers, doctors and therapists read and managed them [77]. Attention to the body is not always benign: past some point, its effects can turn negative [82].

Runaway release (Section 5) is a particular danger for anyone who works forcefully at clearing knots, and no one should face it alone. If it begins, easing off the work that set it off, even if the process itself goes on, the grounding, calming, sleep and exercise that practitioners in difficulty have found helpful [1], and the support of people who know these experiences all belong to getting through it. Any sign of physical illness calls for a physician at once, and drinking large amounts of water is dangerous in itself (Section 3.5). Anyone thinking about suicide should reach out now: in the US, call or text 988; elsewhere, find a line at findahelpline.com.

Nothing in this paper is a practice, a protocol or a recommendation, and nothing in it is a reason to seek these experiences: the hypothesis concerns what happens, not what anyone should do. Anyone in difficulty can find support from Cheetah House [97], whose team includes the researchers of the Varieties of Contemplative Experience project [98], and should also see a physician. Any study with people belongs under ethics review, with ordinary breathing, comfortable pressure and a plan for referral.

10. Conclusion

Knots are among the commonest things a hand finds in a body, and among the least explained. The perforator hypothesis assembles an explanation from parts that are each measured: fixed openings in the fascia; a wall that can hold itself shut, and remembers having been shut; a drive the breath reaches in seconds; a squeeze that widens; a nerve that fires when blood returns; a parent vessel that holds its branches shut. Put together, they describe a knot as a held state of a particular vessel at a particular place, written by the body's drive and kept by the vessel itself. It is, for now, the strongest account in a narrow sense: it explains more of what is reported, with less that is new, and it can be shown wrong quickly: the cheapest test needs an afternoon and a pencil Doppler. If tender points do not sit on perforators more often than chance, or a felt release brings no flood to the skin above it, it should be set aside. If they do, the body keeps some of its history in the calibre of small vessels at fixed openings in its fascia: a register of held states that a hand can find, the breath can reach and an instrument can read.

Box 2. Open questions

- Do tender points sit on perforators more often than chance, and does each person's set of knots recur at the same places?
- Does the skin flood at the moment a knot lets go, and does the flood come before the tingle, and the tingle before the layers slide?
- Can a small human artery held shut for hours stay shut at resting tone?
- What carries a release to attention alone?
- What, if anything, changes in the fascia or the vessels over the days after many knots let go, and what carries runaway release?
- Are the classical points on perforators, or only the tender *ashi* points?

Authorship and AI use

This paper was written by AI under the author's direction. Eugene Jhong conceived the hypothesis, set out first in two essays [2, 4] and developed on the project's site, chose the question and the direction of the argument, and supplied the practitioner observations that motivate it. The text was drafted by Claude Opus 5.5 (Anthropic) from those essays, the site, the project's models and the cited literature. A draft was then refereed by a second model, Claude Fable 5.1 (Anthropic), which checked each claim against its source; its corrections were checked again against the sources before they were made. Every reference was taken from its own record, PubMed's for papers indexed there and the publisher's or Crossref's for the rest, rather than from a model's memory, and every number from the cited sources or the project's data. The author is responsible for the content.

Data and code

The models, every parameter with its source, and the code behind each number on this page are open [8]. The few numbers from the project's own calculations (the wall's switch, the hyaluronan and warming checks, the comparison with acupoints) are read from its data when the page is built.

Competing interests and funding

None.

Versions

Draft 3 (3 November 2026) corrects Box 1: no knot needs months to free; the stubborn ones need more skill (Section 3.3). It adds how a knot forms (Section 3.3); a reported effect of water, which in the young counts against the hypothesis (Section 3.5); runaway release and its dangers (Sections 5 and 9); a reading of the slower phase in the vessels (Section 5); and how the hypothesis stands to vasocomputation, as a case of its idea rather than a rival (the abstract and Sections 1.3, 2, 3.3 and 4). The additions were written by Claude Opus 5.5 under the author's direction and checked by Claude Fable 5.1 against their sources and for safe messaging.

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